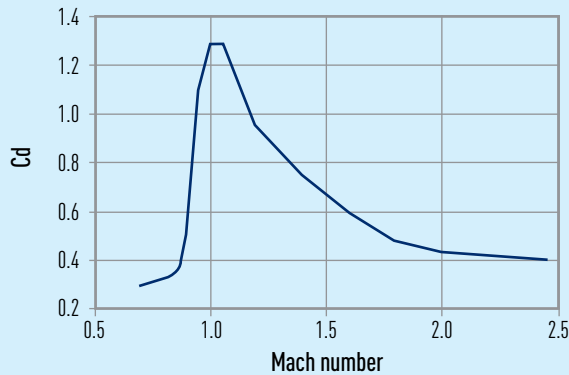


Qualitative variation of Cd with Mach number



Safety First

Commercial flights at best fly at Mach 0.8. Between Mach 0.8 and Mach 1.2, which is known as the transonic range, there is a spike in the amount of drag experienced by the aircraft. It is dangerous to fly passenger aircraft at speeds that are just a little more than current maximum speeds.



Congestion

With an increase in air traffic, most airlines accommodate the congestion and possibility of delays in their flight schedules. Passengers spend more time taxiing on the runway than ever before. With congestion increasing, the scheduled times are also expected to increase.



Development

The industry needs more time to develop safe hypersonic passenger planes, which can travel in excess of five times the speed of sound. It is expected to take another fifty years for hypersonic planes to be safe for passenger travel.



Architecture

Most modern airports are designed to make passengers spend more time in them, which increases the revenues for the stores and restaurants within.



In-flight services

Access to the internet, in flight entertainment services, and seats that expand fully into beds mean that passengers have little incentive to pay exorbitant prices for a faster less comfortable flight.



The Concorde

Two supersonic passenger planes, the **Soviet Tupolev TU 144** and the **Anglo-French Concorde** moved at supersonic speeds, but are no longer in service. The Tupolev TU 144 ceased operations in 1978, while the last commercial flight of the Concorde was in 2003.



The Tupolev TU 144